

HTML Forms

Pemrograman Web 1



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Outline

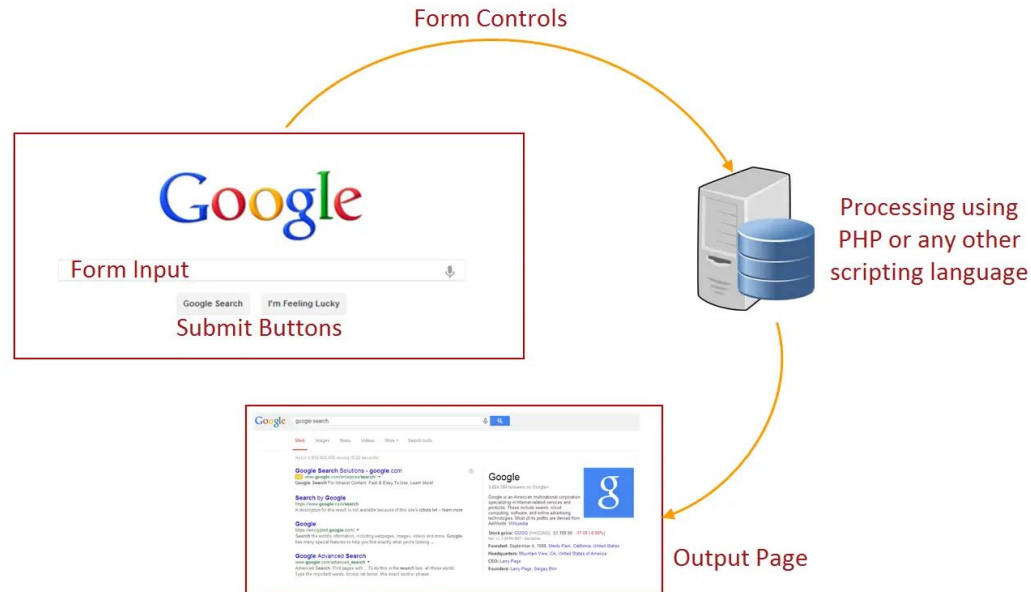
- HTML Forms
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HTML Forms



HTML Forms

An HTML form is used to collect user input. The user input is most often sent to a server for processing.



The <form> Element

The HTML `<form>` element is used to create an HTML form for user input:

```
<form>  
  .  
  form elements  
  .  
</form>
```

The `<form>` element is a container for different types of input elements, such as: text fields, checkboxes, radio buttons, submit buttons, etc.

Form Attributes



The Action Attribute

The **action** attribute defines the action to be performed when the form is submitted.

Usually, the form data is sent to a file on the server when the user clicks on the submit button.

In the example below, the form data is sent to a file called "action_page.php". This file contains a server-side script that handles the form data:

```
<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
  <input type="submit" value="Submit">
</form>
```

The Target Attribute

The **target** attribute specifies where to display the response that is received after submitting the form.

The **target** attribute can have one of the following values:

Value	Description
<code>_blank</code>	The response is displayed in a new window or tab
<code>_self</code>	The response is displayed in the current window
<code>_parent</code>	The response is displayed in the parent frame
<code>_top</code>	The response is displayed in the full body of the window
<code>framename</code>	The response is displayed in a named iframe

```
<form action="/action_page.php" target="_blank">
```

The default value is **_self** which means that the response will open in the current window.

The Method Attribute

The **method** attribute specifies the HTTP method to be used when submitting the form data.

The form-data can be sent as URL variables (with **method="get"**) or as HTTP post transaction (with **method="post"**).

The default HTTP method when submitting form data is GET.

```
<form action="/action_page.php" method="get">
```

```
<form action="/action_page.php" method="post">
```

When to use GET?

The default method when submitting form data is GET.

However, when GET is used, the submitted form data will be visible in the page address field:

/action_page.php?fname=Edo&lname=Riansyah

Notes on GET:

- Appends the form data to the URL, in name/value pairs
- NEVER use GET to send sensitive data! (the submitted form data is visible in the URL!)
- The length of a URL is limited (2048 characters)
- Useful for form submissions where a user wants to bookmark the result
- GET is good for non-secure data, like query strings in Google

When to use POST?

Always use POST if the form data contains sensitive or personal information. The POST method does not display the submitted form data in the page address field.

Notes on POST:

- Appends the form data inside the body of the HTTP request (the submitted form data is not shown in the URL)
- POST has no size limitations, and can be used to send large amounts of data.
- Form submissions with POST cannot be bookmarked

The Autocomplete Attribute

The **autocomplete** attribute specifies whether a form should have autocomplete on or off.

When autocomplete is on, the browser automatically complete values based on values that the user has entered before.

```
<form action="/action_page.php" autocomplete="on">
```

The Novalidate Attribute

The **novalidate** attribute is a boolean attribute.

When present, it specifies that the form-data (input) should not be validated when submitted.

```
<form action="/action_page.php" novalidate>
```

List of All <form> Attributes

Attribute	Description
<u>accept-charset</u>	Specifies the character encodings used for form submission
<u>action</u>	Specifies where to send the form-data when a form is submitted
<u>autocomplete</u>	Specifies whether a form should have autocomplete on or off
<u>enctype</u>	Specifies how the form-data should be encoded when submitting it to the server (only for method="post")
<u>method</u>	Specifies the HTTP method to use when sending form-data
<u>name</u>	Specifies the name of the form
<u>novalidate</u>	Specifies that the form should not be validated when submitted
<u>rel</u>	Specifies the relationship between a linked resource and the current document
<u>target</u>	Specifies where to display the response that is received after submitting the form

Form Elements



The HTML <form> Elements

The HTML `<form>` element can contain one or more of the following form elements:

- `<input>`
- `<label>`
- `<select>`
- `<textarea>`
- `<button>`
- `<fieldset>`
- `<legend>`
- `<datalist>`
- `<output>`
- `<option>`
- `<optgroup>`



The <input> Element

One of the most used form elements is the `<input>` element.

The `<input>` element can be displayed in several ways, depending on the `type` attribute.

```
<label for="fname">First name:</label>  
<input type="text" id="fname" name="fname">
```

The <label> Element

The `<label>` element defines a label for several form elements.

The `<label>` element is useful for screen-reader users, because the screen-reader will read out loud the label when the user focus on the input element.

The `<label>` element also help users who have difficulty clicking on very small regions (such as radio buttons or checkboxes) - because when the user clicks the text within the `<label>` element, it toggles the radio button/checkbox.

The `for` attribute of the `<label>` tag should be equal to the `id` attribute of the `<input>` element to bind them together.

The <select> Element

The <select> element defines a drop-down list:

```
<label for="cars">Choose a car:</label>
<select id="cars" name="cars">
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

The <option> element defines an option that can be selected.

By default, the first item in the drop-down list is selected.

To define a pre-selected option, add the **selected** attribute to the option:

```
<option value="fiat" selected>Fiat</option>
```

<select> Element - Allow Multiple Selections

Use the **multiple** attribute to allow the user to select more than one value:

```
<label for="cars">Choose a car:</label>
<select id="cars" name="cars" size="4" multiple>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

The <textarea> Element

The <textarea> element defines a multi-line input field (a text area):

```
<textarea name="message" rows="10" cols="30">  
The cat was playing in the garden.  
</textarea>
```

The **rows** attribute specifies the visible number of lines in a text area.

The **cols** attribute specifies the visible width of a text area.

The <button> Element

The `<button>` element defines a clickable button:

```
<button type="button" onclick="alert('Hello World!')">Click Me!</button>
```

The <fieldset> and <legend> Elements

The <fieldset> element is used to group related data in a form.

The <legend> element defines a caption for the <fieldset> element.

```
<form action="/action_page.php">
  <fieldset>
    <legend>Personalia:</legend>
    <label for="fname">First name:</label><br>
    <input type="text" id="fname" name="fname" value="John"><br>
    <label for="lname">Last name:</label><br>
    <input type="text" id="lname" name="lname" value="Doe"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```

The <datalist> Element

The `<datalist>` element specifies a list of pre-defined options for an `<input>` element.

Users will see a drop-down list of the pre-defined options as they input data.

The list attribute of the `<input>` element, must refer to the id attribute of the `<datalist>` element.

```
<form action="/action_page.php">
  <input list="browsers">
    <datalist id="browsers">
      <option value="Edge">
      <option value="Firefox">
      <option value="Chrome">
      <option value="Opera">
      <option value="Safari">
    </datalist>
  </form>
```


Input Types



Input Type Text

`<input type="text">` defines a **single-line text input field**:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname">
</form>
```

First name:

Last name:

Input Type Password

`<input type="password">` defines a **password field**:

```
<form>
  <label for="username">Username:</label><br>
  <input type="text" id="username" name="username"><br>
  <label for="pwd">Password:</label><br>
  <input type="password" id="pwd" name="pwd">
</form>
```

Username:

Password:

Input Type Submit

`<input type="submit">` defines a button for **submitting** form data to a **form-handler**.

The form-handler is typically a server page with a script for processing input data.

The form-handler is specified in the form's **action** attribute:

```
<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
  <input type="submit" value="Submit">
</form>
```

First name:

Last name:

Input Type Reset

`<input type="reset">` defines a **reset button** that will reset all form values to their default values:

```
<form action="/action_page.php">
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe"><br><br>
  <input type="submit" value="Submit">
  <input type="reset" value="Reset">
</form>
```

First name:

John

Last name:

Doe

Submit

Reset

Input Type Radio

`<input type="radio">` defines a **radio button**.

Radio buttons let a user select ONLY ONE of a limited number of choices:

```
<p>Choose your favorite Web language:</p>

<form>
  <input type="radio" id="html" name="fav_language" value="HTML">
  <label for="html">HTML</label><br>
  <input type="radio" id="css" name="fav_language" value="CSS">
  <label for="css">CSS</label><br>
  <input type="radio" id="javascript" name="fav_language" value="JavaScript">
  <label for="javascript">JavaScript</label>
</form>
```

- ☐ HTML
- ☐ CSS
- ☐ JavaScript

Input Type Checkbox

`<input type="checkbox">` defines a **checkbox**.

Checkboxes let a user select ZERO or MORE options of a limited number of choices.

```
<form>
  <input type="checkbox" id="vehicle1" name="vehicle1" value="Bike">
  <label for="vehicle1"> I have a bike</label><br>
  <input type="checkbox" id="vehicle2" name="vehicle2" value="Car">
  <label for="vehicle2"> I have a car</label><br>
  <input type="checkbox" id="vehicle3" name="vehicle3" value="Boat">
  <label for="vehicle3"> I have a boat</label>
</form>
```

- ☐ I have a bike
- ☐ I have a car
- ☐ I have a boat

Input Type Button

`<input type="button">` defines a **button**:

```
<input type="button" onclick="alert('Hello World!')" value="Click Me!">
```


Input Type Color

The `<input type="color">` is used for input fields that should contain a color.

Depending on browser support, a color picker can show up in the input field.

```
<form>
  <label for="favcolor">Select your favorite color:</label>
  <input type="color" id="favcolor" name="favcolor">
</form>
```

Select your favorite color:

Input Type Date

The `<input type="date">` is used for input fields that should contain a date.

Depending on browser support, a date picker can show up in the input field.

```
<form>
  <label for="birthday">Birthday:</label>
  <input type="date" id="birthday" name="birthday">
</form>
```

Birthday:

Input Type File

The `<input type="file">` defines a file-select field and a "Browse" button for file uploads.

```
<form>
  <label for="myfile">Select a file:</label>
  <input type="file" id="myfile" name="myfile">
</form>
```

Select a file: No file chosen

HTML Input Types

Here are the different input types you can use in HTML:

- `<input type="button">`
- `<input type="checkbox">`
- `<input type="color">`
- `<input type="date">`
- `<input type="datetime-local">`
- `<input type="email">`
- `<input type="file">`
- `<input type="hidden">`
- `<input type="image">`
- `<input type="month">`
- `<input type="number">`
- `<input type="password">`
- `<input type="radio">`
- `<input type="range">`
- `<input type="reset">`
- `<input type="search">`
- `<input type="submit">`
- `<input type="tel">`
- `<input type="text">`
- `<input type="time">`
- `<input type="url">`
- `<input type="week">`

Input Attributes



The value Attribute

The input **value** attribute specifies an initial value for an input field:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe">
</form>
```

First name:

Last name:

The readonly Attribute

The input **readonly** attribute specifies that an input field is read-only.

A read-only input field cannot be modified (however, a user can tab to it, highlight it, and copy the text from it).

The value of a read-only input field will be sent when submitting the form!

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John" readonly><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe">
</form>
```

The disabled Attribute

The input **disabled** attribute specifies that an input field should be disabled.

A disabled input field is unusable and un-clickable.

The value of a disabled input field will not be sent when submitting the form!

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" value="John" disabled><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname" value="Doe">
</form>
```


The size Attribute

The input **size** attribute specifies the visible width, in characters, of an input field.

The default value for **size** is 20.

Note: The **size** attribute works with the following input types: text, search, tel, url, email, and password.

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" size="50"><br>
  <label for="pin">PIN:</label><br>
  <input type="text" id="pin" name="pin" size="4">
</form>
```

The maxlength Attribute

The input **maxlength** attribute specifies the maximum number of characters allowed in an input field.

Note: When a **maxlength** is set, the input field will not accept more than the specified number of characters. However, this attribute does not provide any feedback. So, if you want to alert the user, you must write JavaScript code.

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" size="50"><br>
  <label for="pin">PIN:</label><br>
  <input type="text" id="pin" name="pin" maxlength="4" size="4">
</form>
```

The placeholder Attribute

The input **placeholder** attribute specifies a short hint that describes the expected value of an input field (a sample value or a short description of the expected format).

The short hint is displayed in the input field before the user enters a value.

The **placeholder** attribute works with the following input types: text, search, url, number, tel, email, and password.

```
<form>
  <label for="phone">Enter a phone number:</label>
  <input type="tel" id="phone" name="phone"
    placeholder="123-45-678"
    pattern="[0-9]{3}-[0-9]{2}-[0-9]{3}">
</form>
```

Enter a phone number:

The required Attribute

The input **required** attribute specifies that an input field must be filled out before submitting the form.

The **required** attribute works with the following input types: text, search, url, tel, email, password, date pickers, number, checkbox, radio, and file.

```
<form>
  <label for="username">Username:</label>
  <input type="text" id="username" name="username" required>
</form>
```

The autofocus Attribute

The input **autofocus** attribute specifies that an input field should automatically get focus when the page loads.

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text" id="fname" name="fname" autofocus><br>
  <label for="lname">Last name:</label><br>
  <input type="text" id="lname" name="lname">
</form>
```

Referensi:

- W3Schools - HTML Tutorial:
<https://www.w3schools.com/html/default.asp>
- W3Schools - HTML Tutorial Playlist:
https://www.youtube.com/playlist?list=PLP9IO4UYNF0VdAajP_5pYG-jG2JRrG72s
- HTML Tutorial - GeeksforGeeks:
<https://www.geeksforgeeks.org/html/html-tutorial/>
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